

Frank Schlehofer

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EDUCATION

Purdue University, West Lafayette, IN

Graduation May 2027

Bachelor of Science in Computer Science | **CGPA: 3.98/4.0**

SKILLS

Programming: Java, C, C++, Python, JavaScript, HTML, CSS

Frameworks/Tools: React, Node, Express, PostgreSQL, Git, OpenCV, TensorFlow, PyTorch, Pandas, Scikit-Learn

Familiar With: AWS, Swift, x86 Assembly, Google Cloud Vision API

PROJECTS

Full Stack Fridge Management Web App | Node, React, Express, REST API, Javascript March 2025 - Present

- Developed a Node.js and Express backend server with RESTful API endpoints for managing user data.
- Built a React.js frontend styled with CSS to create a responsive and intuitive user interface.
- Integrated PostgreSQL for persistent data storage with scalable, modular code architecture.
- Practiced full agile methodology with weekly sprints and version control via GitHub.
- Connected with both Google Cloud Vision and Spoonacular API for additional features.

Hybrid Movie Recommendation System | Python, PyTorch, Pandas, Scikit-Learn, Matplotlib June 2025

- Developed a collaborative filtering model in PyTorch to predict movie ratings on the MovieLens dataset.
- Engineered a complete ML pipeline, including exploratory data analysis, data preprocessing, and batch loading using PyTorch Dataset and DataLoader.
- Systematically diagnosed and solved model performance issues, including overfitting and unconstrained outputs, improving the validation RMSE from over 5.0 to 1.05.
- Conducted hyperparameter tuning experiments for learning rate and model complexity to optimize performance.

Handwritten Number Detection Model | Python, OpenCV, Tensorflow December 2024

- Used OpenCV to detect and track hand and finger positions with a webcam.
- Trained a simple machine learning model with TensorFlow using handwritten numbers in the MNIST data set.

AWS Open-Source Contributor Initiative | AWS, Typescript, Git September 2023 – December 2023

- Developed an Open Search Dashboards plugin called “Search Relevance” that improved the relevance ranking of search results.
- Collaborated with an agile development team using GitHub, AWS services, and JavaScript/TypeScript to enhance user search experiences.

EXPERIENCE

Undergraduate Teaching Assistant, Purdue University

Jan 2025 – Present

Foundations of C

- Led weekly lab sessions to reinforce fundamental C programming concepts.
- Assisted students with debugging, memory management, and algorithm design.
- Provided individualized feedback on assignments to improve coding practices and problem-solving skills.

RELEVANT COURSEWORK

Data Structures and Algorithms, Systems Programming, Object-Oriented Programming, Discrete Math, C Programming, Computer Architecture, Statistics, Probability, Artificial Intelligence, Analysis of Algorithms, Calculus II & III, Linear Algebra

